

It worked perfectly. Then: traffic.

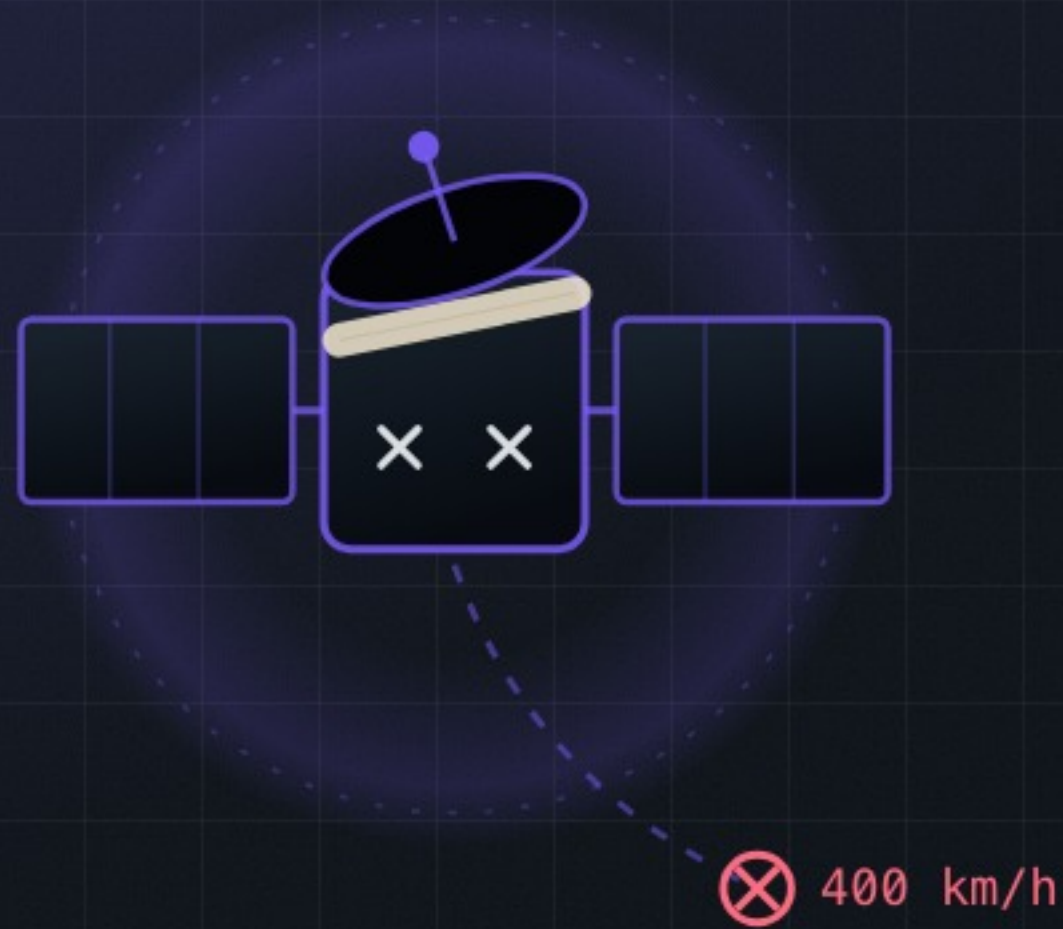
Our jitter filter was flawless
right up until someone sat in
Bangalore traffic.

It could not tell a parked car
from one moving at walking
pace.

5m



One number, every context



THE CONCUSSED WITNESS

exhibit 01

LocationManager

A single global threshold is the most confident wrong decision in your codebase. It swears the same thing is true while walking, cycling, driving, and standing still.



The Loopdown

@darkpandawarrior

02 / 05

swipe

// THE CLAIM YOU MADE

A constant is an assumption

```
private fun minDisplacementForSpeed(speedMps: Double): Double = when {  
    speedMps < 2.5 -> 2.0    // walking  
    speedMps < 7.0 -> 3.0    // cycling  
    else           -> 5.0    // driving  
}
```



Thresholds take context

- 01 Function of speed band**
walking and motorway are not the same filter
- 02 Function of time since last fix**
a 30 second gap changes what is plausible
- 03 Function of recent history**
what was this device doing ten seconds ago
- 04 Name the assumption in code**
if it is "all contexts", that is the bug



A constant with no context is a bug on a delay

When you write a number into a filter, ask which context you are assuming.

If the answer is "all of them", you have not written a threshold.